

Density Based Traffic Light Control System

Project Report submitted to

**Chhattisgarh Swami Vivekanand Technical University
Bhilai(India)**

In partial fulfillment for award of the degree of

BACHELOR OF ENGINEERING

In

Electrical and Electronics Engineering

by

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DECLARATION BY THE STUDENT'S

We the undersigned solemnly declare that the project report title “ **Density based Traffic Light Control System**” is based on our own work carried out during the course of our study under the supervision of Mrs.Naushin Anjum.

We assert that the statements made and conclusions drawn are an outcome of our work. We further certify that

- i. The work contained in the report is original and has been done by us under the general supervision of our supervisor(s).
- ii. The work has not been submitted to any other Institute for any other degree/diploma/certificate in this university or any other University of India or abroad.
- iii. We have followed the guidelines provided by the University in writing the report.
- iv. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references.

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the work incorporated in the project report entitled “**Density based Traffic Light control System**” is a record of work carried out by Anshul Lanjewar bearing Enrollment No.BH4679, Animesh Sinha bearing Enrollment No.BH4676. , Stuti Mishra bearing Enrollment No.:BH4722, Rishita Upadhyay bearing Enrollment No BH4695 and Sanskriti Gauraha bearing Enrollment No BH4709 under my/our guidance and supervision for the award of Degree of Bachelor of Engineering in the faculty of Department of Electrical and Electronics Engineering of Chhattisgarh Swami Vivekanand Technical University, Bhilai, Chhattisgarh, India.

To the best of my/our knowledge and belief the project report

- i) Embodies the work of the candidates themselves,
- ii) Has duly been completed,
- iii) Fulfills the requirement BE degree of the University and
- iv) Is up to the desired standard both in respect of contents and language for being referred to the examiners.

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ABSTRACT

At present, vehicular traffic is increasing across the world, especially in large urban areas. As the number of road users is increasing rapidly and current resources & infrastructures being limited; a smart traffic control will become a very important issue in the future. These needs have led to an ever increasing demand for an “intelligent” traffic control system. Therefore, optimization of traffic control to efficiently and smoothly accommodate this increasing demand is needed. Our project will demonstrate the optimization of traffic lights in a city using wireless sensors. Traffic light optimization is a tough problem. With multiple junctions, the complexity increases as the state of one light node influences the flow of traffic towards many other nodes. We proposed a traffic light controller that allows us to control and study different situations of traffic density. We sense the density of traffic using infra-red sensors. The key role behind the implementation of the “Traffic density based light control system” is to make use of an AT89C51 controller which performs processing of the real time data provided by the infra-red sensors, eventually controlling the traffic flow via the LED traffic lights. With this detection technique traffic can be managed and controlled more effortlessly and smartly.

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CHAPTER-1

INTRODUCTION

1.1 Overview

In the past, the researchers had gone through different types of technologies. In today's high speed life, traffic congestion becomes a serious issue in our day to day activities. It brings down the productivity of individuals and thereby the society as lots of work hours are wasted in the signals. High volume of vehicles, the inadequate infrastructure and the irrational distribution of the signaling system are main reasons for this chaotic congestion. It indirectly also adds to the increase in pollution level as engines remain on in most cases, a huge volume of natural resources in forms of petrol and diesel is consumed without any fruitful outcome. Therefore, in order to get rid of these problems or at least reduce them to a significant level, newer schemes need to be implemented by bringing in sensor based automation techniques in this field of traffic signaling systems. intricate reasoning while deciding when to change the lights for the various road users waiting in different lanes. How long the signal stays green in one lane and red in another is most often determined by simple timing that is calculated when the crossing is designed. Even though today's methods are robust and work well when the traffic load is distributed evenly across the lanes in the intersection, the system turns out to be inefficient when there are complications in the density of traffic which in turn results in chaos. Unnecessary waiting time in the signal can be avoided by determining in which side the green signal should be large during the traffic. This research is to design such a system which works on the traffic density and manages the signal lights according to the sensed density of the traffic through the infra-red Sensors. The timing of the signal lights will vary with respect to the varying density of the traffic, Hence Improving the light system and reducing the traffic congestion and other related problems. It indirectly also adds to the increase in pollution level as engines remain on in most cases, a huge volume of natural resources in forms of petrol and diesel is consumed without any fruitful outcome. Therefore, in order to get rid of these problems or at least reduce them to a significant level, newer schemes need to be implemented by bringing in sensor based automation techniques in this field of traffic signaling system.

1.2 Objective

The main purpose of introducing this smart traffic system is done so that every time when the vehicles at a junction become dense and the traffic lights get changed, calculate at each side for some fixed time. Even though there are no vehicles at a particular side, the traffic signals will glow for a given fixed time. Due to that there is time wastage & vehicles on the other side have to wait for the time to complete the process. So to reduce the wastage of time, and to avoid problems related to congestion, we can implement a system that controls the traffic based on the heavy flow of vehicles at any particular side. With this system, we shall count the number of vehicles at each side at the junction and give a path to the particular side which has denser traffic and keeping the other sides stopped. Our project aims at reducing traffic congestion and unwanted long time delay during the traffic light switch covers especially when the traffic is very low. It is designed to be implemented in places nearing the junctions where the traffic signals are placed, in order to reduce the congestion in these junctions. It keeps a track of the vehicles in each road and accordingly adjusts the time for each traffic light signal. The higher the number of vehicles on the road the longer will be the time delay allotted for that corresponding traffic light signal. The main purpose of this project is, if there will be no traffic on the other signal, one shouldn't wait for that signal. The system will skip that signal and will move on the next one

The project is a traffic signal controller system that reacts to the density of traffic. The time required on changing signals adjusts automatically depending on the density of traffic. Well traffic problems contribute to a major problem in many cities and also increase fuel consumption as well as pollution. Traffic signals operating on fixed signal timing delays do not adjust to the changing traffic density. When the traffic density increases more than a limit at one particular side, it needs a longer green light duration to ease traffic flow. Our system uses a microcontroller of the 8051 family that is interfaced with IR sensors. These IR sensors are used for line of sight object detection using which the system gets an input of the traffic density. Traffic density is measured as "low, medium and high". Based on this density reading the system adjusts and varies the traffic signal duration for that particular way. This system can also be further enhanced by using multiple such systems across the city and synchronizing them through the network. So, in this paper we are going to focus on the arrangement of traffic lights because static arrangements have a fixed time allocation at each junction and it's not able to perform according to density of traffic available at the junction. So density based traffic light control can be considered a solution to reduce traffic congestion in some possible way at the road. In density based traffic light control sensors and microcontrollers play a major role to sense the traffic and transfer the information to the next level for receiving an output. The main objective is to focus on this problem only to reduce traffic congestion, wastage of fuel, and collisions due to the high number of vehicles at the time of traveling.

1.3 Problem Statement

Traffic congestion is increasing on the road day- by- day. As a result of which, two main issues arise. First is congestion due to dense traffic on the red side which results in jams and accidents and other is unnecessary wait time when there is no traffic on the green light side. These problems occur due to fixed control on traffic. This research will aim to control the traffic wirelessly according to the density, but in a manner of programming which is already fixed in the system. There are many problems with the conventional traffic light controller. One of them is heavy traffic jammed where there has not been any realization on how to measure the level of this „jammed “ thus pounder on the solution especially using time delay. Another problem with conventional traffic lights are when there is no traffic, but the waiting still continues . Both Development of the economy and increasing population are factors contributing to traffic congestions. Traffic congestion is a condition on road networks that happened as the road users increased thus leading to slower speeds, longer trip times, and increased queuing time .A traffic violation detection system must be realized in real time as the authorities track the roads all the time. Hence, traffic enforcers will not only be at ease in implementing safe roads accurately but also efficiently; as the traffic detection system detects violations faster than humans.

CHAPTER-2

LITERATURE REVIEW

Currently the vehicle problem is increasing exponentially and Traffic congestion is a severe problem in many modern cities all over the world. To overcome the problem, we have come up with the idea of Density based traffic light control system. Traffic research has the goal to optimize traffic flow, as roads have been overloaded with the increasing number of vehicles. There are several models which give solutions for traffic simulation. In our research we have focused on controlling the traffic lights on the basis of traffic density. Nowadays congestion in traffic is a serious issue. The traffic congestion can also be caused by large Red light delays, etc. The delay of each light is hard coded in the traffic light and it is not dependent on traffic. Thus we propose multiple traffic light control systems. The system tries to reduce possibilities of traffic jams, caused by traffic lights, up to a great extent. The system is based on a microcontroller. The microcontroller used in the system is AT89C51. The system contains IR transmitter and IR receiver which are mounted on either side of roads respectively. The IR system gets activated whenever any vehicle passes on road between IR transmitter and IR receiver. Microcontroller controls the IR system and counts the number of vehicles passing on the road. Microcontrollers also store vehicles' count in its memory. Based on different vehicle counts, the microcontroller takes decisions and updates the traffic light delays as a result. The traffic light is situated at a certain distance from the IR system. Thus based on vehicle count, microcontrollers define different ranges for traffic light delays and update those accordingly. In this system IR sensors are used to measure the density of the vehicles which are fixed within a fixed distance. All the sensors are interfaced with the microcontroller which in turn controls the traffic signals system according to density detected by the sensors. If the traffic density is high on a particular side more priority is given for that side .. The sensors continuously keep sensing density on all sides and the green signal is given on priority basis, where the sensors detect high density. The side with the next priority level follows the first priority level. By using this system traffic can be cleared without irregularities and time delay when there is no traffic on the other side can be avoided

CHAPTER -3

METHODOLOGY(Structure And Operation)

3.1 Components Used:

3.1.1 ARDUINO ATmega328

Arduino is an open-source hardware and software company, project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its hardware products are licensed under a CC BY-SA license, while the software is licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially from the official website or through authorized distributors.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards ('shields') or breadboards (for prototyping) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs. The microcontrollers can be programmed using the C and C++ programming languages, using a standard API which is also known as the **Arduino Programming Language**, inspired by the Processing language and used with a modified version of the Processing IDE. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) and a command line tool developed in Go.

The Arduino project began in 2005 as a tool for students at the Interaction Design Institute Ivrea, Italy, aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion

detectors. The name *Arduino* comes from a bar in Ivrea, Italy, where some of the project's founders used to meet. The bar was named after Arduin of Ivrea, who was the margrave of the March of Ivrea and King of Italy from 1002 to 1014.

Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available.

Most Arduino boards consist of an Atmel 8-bit AVR microcontroller (ATmega8, ATmega168, ATmega328, ATmega1280, or ATmega2560) with varying amounts of flash memory, pins, and features. The 32-bit Arduino Due, based on the Atmel SAM3X8E was introduced in 2012. The boards use single or double-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed *shields*. Multiple and possibly stacked shields may be individually addressable via an I²C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator. Some designs, such as the LilyPad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.

Arduino microcontrollers are pre-programmed with a boot loader that simplifies the uploading of programs to the on-chip flash memory. The default bootloader of the Arduino Uno is the Optiboot bootloader. Boards are loaded with program code via a serial connection to another computer. Some serial Arduino boards contain a level shifter circuit to convert between RS-232 logic levels and transistor–transistor logic (TTL) level signals. Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232. Some boards, such as later-model Uno boards, substitute the FTDI chip with a separate AVR chip containing USB-to-serial firmware, which is reprogrammable via its own ICSP header. Other variants, such as the Arduino Mini and the unofficial Boarduino, use a detachable USB-to-serial adapter board or cable, Bluetooth or other methods. When used with traditional microcontroller tools, instead of the Arduino IDE, standard AVR in-system programming (ISP) programming is used.

The ATmega328P is the microcontroller that powers the Arduino Uno development board. The Arduino board makes it easy to interface with the pins on the ATmega328P while adding extra

features that don't come with the standalone microcontroller, including a USB serial interface and 16 MHz clock.

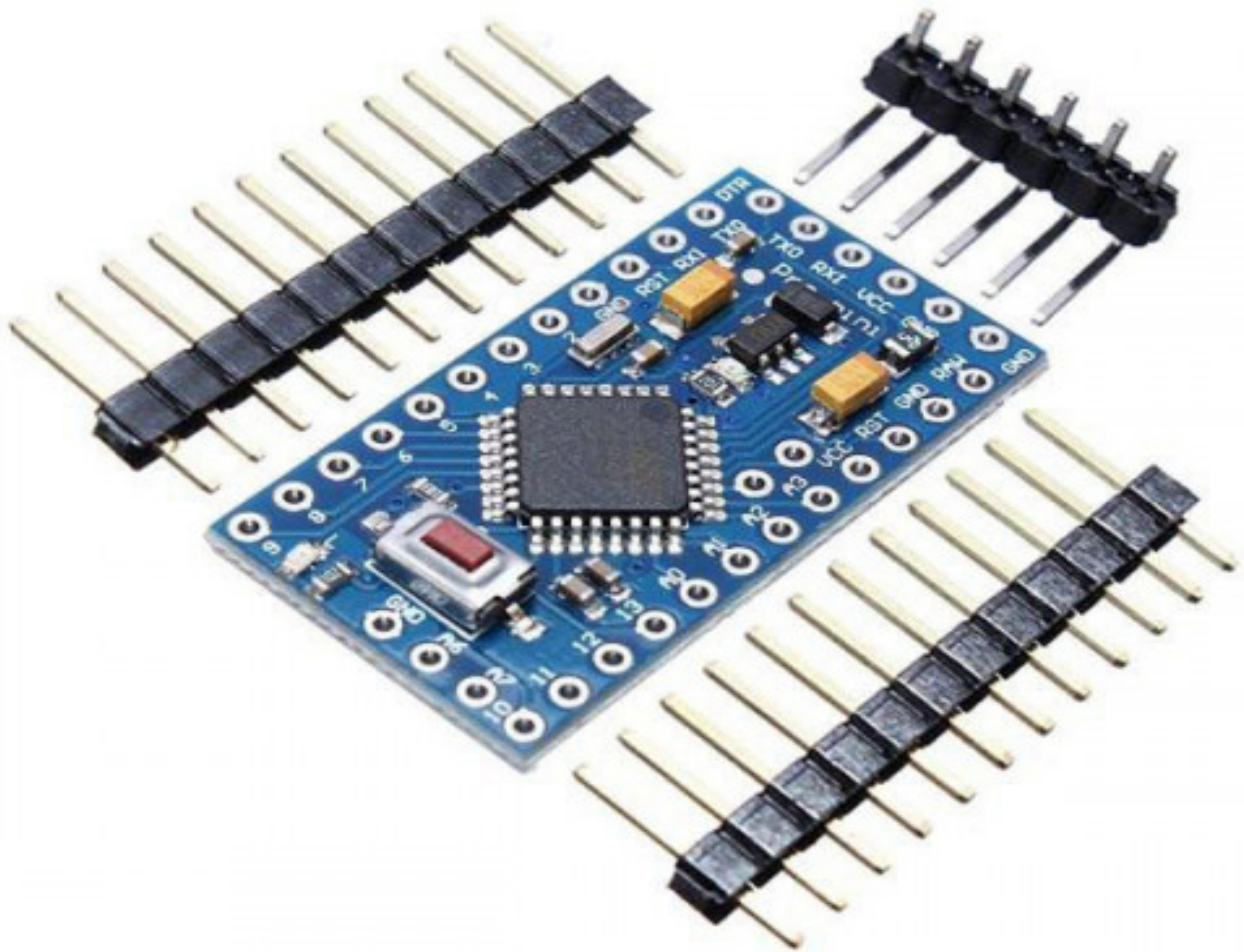


Fig 3.1. ARDUINO ATmega328P

3.1.2 TRANSFORMER (12-0-12)

A **transformer** is a passive component that transfers electrical energy from one electrical circuit to another circuit, or multiple circuits. A varying current in any coil of the transformer produces

a varying magnetic flux in the transformer's core, which induces a varying electromotive force (EMF) across any other coils wound around the same core. Electrical energy can be transferred between separate coils without a metallic (conductive) connection between the two circuits. Faraday's law of induction, discovered in 1831, describes the induced voltage effect in any coil due to a changing magnetic flux encircled by the coil.

Transformers are used to change AC voltage levels, such transformers being termed step-up or step-down type to increase or decrease voltage level, respectively. Transformers can also be used to provide galvanic isolation between circuits as well as to couple stages of signal-processing circuits. Since the invention of the first constant-potential transformer in 1885, transformers have become essential for the transmission, distribution, and utilization of alternating current electric power. A wide range of transformer designs is encountered in electronic and electric power applications. Transformers range in size from RF transformers less than a cubic centimeter in volume, to units weighing hundreds of tons used to interconnect the power grid.

Various specific electrical application designs require a variety of transformer types. Although they all share the basic characteristic transformer principles, they are customized in construction or electrical properties for certain installation requirements or circuit conditions. In electric power transmission, transformers allow transmission of electric power at high voltages, which reduces the loss due to heating of the wires. This allows generating plants to be located economically at a distance from electrical consumers. All but a tiny fraction of the world's electrical power has passed through a series of transformers by the time it reaches the consumer. In many electronic devices, a transformer is used to convert voltage from the distribution wiring to convenient values for the circuit requirements, either directly at the power line frequency or through a switch mode power supply.

Signal and audio transformers are used to couple stages of amplifiers and to match devices such as microphones and record players to the input of amplifiers. Audio transformers allowed telephone circuits to carry on a two-way conversation over a single pair of wires. A balun transformer converts a signal that is referenced to ground to a signal that has balanced voltages to ground, such as between external cables and internal circuits. Isolation transformers prevent leakage of current into the secondary circuit and are used in medical equipment and at

construction sites. Resonant transformers are used for coupling between stages of radio receivers, or in high-voltage Tesla coils.

1A center- tapped step down transformer is a general -purpose chassis mounting mains transformer .Transformer has 230v primary winding and center-tapped secondary winding . The transformer acts as a step down transformer reducing 230VAC RMS to 12VAC RMS .

Specification:-

1. Step down transformer
2. Input voltage: 230V AC
3. Output voltage : 12V, 0V or ground and 12V
4. Output current: 3A
5. Max. frequency range: 50Hz to 60Hz
6. Type: Vertical mount type
7. High-quality copper wire winding
8. Long service life with excellent performance



Fig 3.2. TRANSFORMER (12-0-12)

3.1.3 7-segment display

A seven-segment display is a form of electronic display device for displaying decimal numerals that is an alternative to the more complex dot matrix displays. Seven-segment displays are widely used in digital clocks, electronic meters, basic calculators, and other electronic devices that display numerical information.

Seven-segment displays may use a liquid crystal display (LCD), a light-emitting diode (LED) for each segment, an electrochromic display, or other light-generating or controlling techniques such as cold cathode gas discharge (Panaplex), vacuum fluorescent (VFD), incandescent filaments (Numitron), and others. For gasoline price totems and other large signs, vane displays made up of electromagnetically flipped light-reflecting segments (or "vanes") are still commonly used. A precursor to the 7-segment display in the 1950s through the 1970s was the cold-cathode, neon-lamp-like nixie tube. Starting in 1970, RCA sold a display device known as the *Numitron* that used incandescent filaments arranged into a seven-segment display. In the USSR, the first electronic calculator "Vega", which was produced from 1964, contained 20 decimal digits with a seven-segment electroluminescent display.

In a simple LED package, typically all of the cathodes (negative terminals) or all of the anodes (positive terminals) of the segment LEDs are connected and brought out to a common pin; this is referred to as a "common cathode" or "common anode" device. Hence a 7 segment plus decimal point package will only require nine pins, though commercial products typically contain more pins, and/or spaces where pins would go, in order to match standard IC sockets. Integrated displays also exist, with single or multiple digits. Some of these integrated displays incorporate their own internal decoder, though most do not: each individual LED is brought out to a connecting pin as described.

Multiple-digit LED displays as used in pocket calculators and similar devices used multiplexed displays to reduce the number of I/O pins required to control the display. For example, all the anodes of the A segments of each digit position would be connected together and to a driver circuit pin, while the cathodes of all segments for each digit would be connected. To operate any particular segment of any digit, the controlling integrated circuit would turn on the cathode driver for the selected digit, and the anode drivers for the desired segments; then after a short

blanking interval the next digit would be selected and new segments lit, in a sequential fashion. In this manner an eight digit display with seven segments and a decimal point would require only 8 cathode drivers and 8 anode drivers, instead of sixty-four drivers and IC pins. Often in pocket calculators the digit drive lines would be used to scan the keyboard as well, providing further savings; however, pressing multiple keys at once would produce odd results on the multiplexed display.

Although to a naked eye all digits of an LED display appear lit, only one digit is lit at any given time in a multiplexed display. The digit changes at a high enough rate that the human eye cannot see the flashing (on earlier devices it could be visible to peripheral vision).

A seven-segment LED is a digital display module specialized to display numerical information. Light-emitting diodes (LEDs) arranged in the shape of numbers offer an easily visible display. They are sometimes called "seven-segment displays" or "seven-segment indicators."

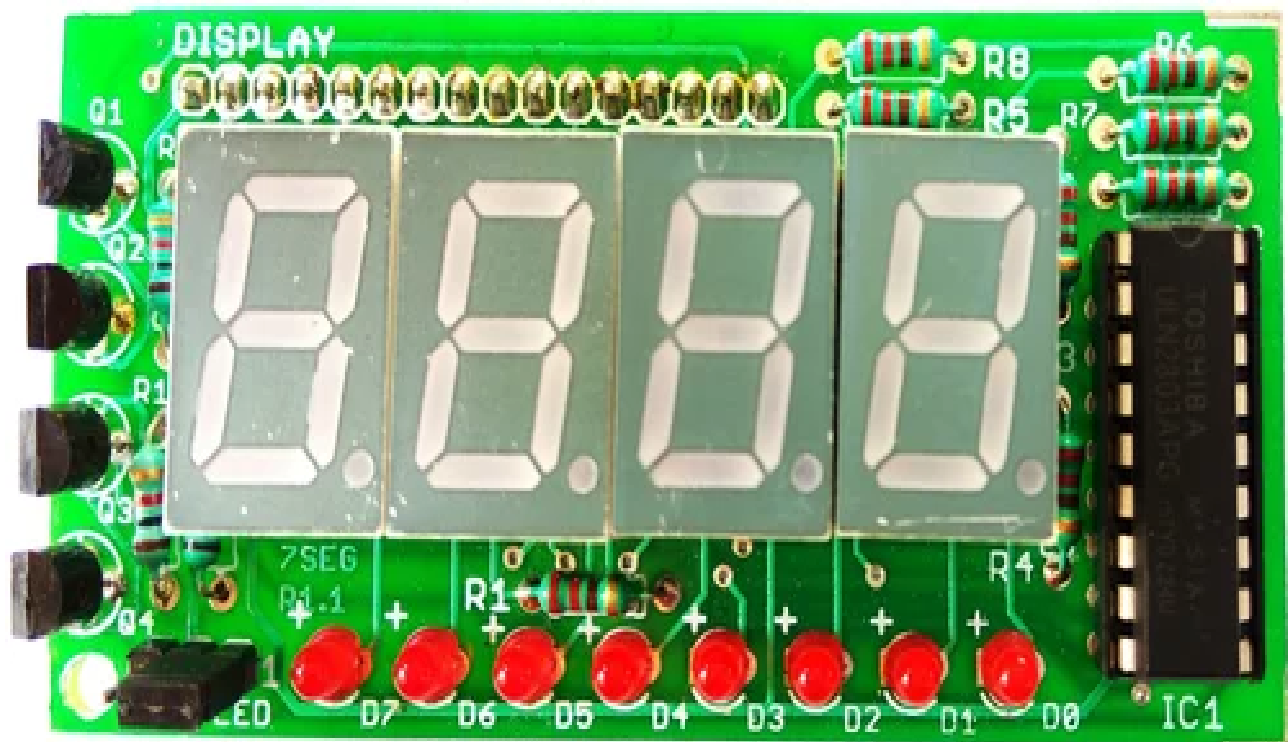


Fig 3.3. 7-segment display

3.1.4 CAPACITOR

A **capacitor** is a device that stores electrical energy in an electric field by virtue of accumulating electric charges on two close surfaces insulated from each other. It is a passive electronic component with two terminals.

The effect of a capacitor is known as capacitance. While some capacitance exists between any two electrical conductors in proximity in a circuit, a capacitor is a component designed to add capacitance to a circuit. The capacitor was originally known as the **condenser**, a term still encountered in a few compound names, such as the *condenser microphone*.

The physical form and construction of practical capacitors vary widely and many types of capacitor are in common use. Most capacitors contain at least two electrical conductors often in the form of metallic plates or surfaces separated by a dielectric medium. A conductor may be a foil, thin film, sintered bead of metal, or an electrolyte. The nonconducting dielectric acts to increase the capacitor's charge capacity. Materials commonly used as dielectrics include glass, ceramic, plastic film, paper, mica, air, and oxide layers. Capacitors are widely used as parts of electrical circuits in many common electrical devices. Unlike a resistor, an ideal capacitor does not dissipate energy, although real-life capacitors do dissipate a small amount (see Non-ideal behavior). When an electric potential difference (a voltage) is applied across the terminals of a capacitor, for example when a capacitor is connected across a battery, an electric field develops across the dielectric, causing a net positive charge to collect on one plate and net negative charge to collect on the other plate. No current actually flows through the dielectric. However, there is a flow of charge through the source circuit. If the condition is maintained sufficiently long, the current through the source circuit ceases. If a time-varying voltage is applied across the leads of the capacitor, the source experiences an ongoing current due to the charging and discharging cycles of the capacitor.

The earliest forms of capacitors were created in the 1740s, when European experimenters discovered that electric charge could be stored in water-filled glass jars that came to be known as Leyden jars. Today, capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass. In analog filter networks, they smooth the output of power supplies. In resonant circuits they tune radios to particular frequencies. In electric power

transmission systems, they stabilize voltage and power flow. The property of energy storage in capacitors was exploited as dynamic memory in early digital computers, and still is in modern DRAM. Device for storing electrical energy, consisting of two conductors in close proximity and insulating from each other. It is used for converting pulsating DC into pure DC supply.



Fig 3.4. CAPACITOR

3.1.5 VOLTAGE REGULATOR:

A **voltage regulator** is a system designed to automatically maintain a constant voltage. A voltage regulator may use a simple feed-forward design or may include negative feedback. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or DC voltages.

Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line.

A simple voltage/current regulator can be made from a resistor in series with a diode (or series of diodes). Due to the logarithmic shape of diode V-I curves, the voltage across the diode changes only slightly due to changes in current drawn or changes in the input. When precise voltage control and efficiency are not important, this design may be fine. Since the forward voltage of a diode is small, this kind of voltage regulator is only suitable for low voltage regulated output. When higher voltage output is needed, a zener diode or series of zener diodes may be employed. Zener diode regulators make use of the zener diode's fixed reverse voltage, which can be quite large.

Feedback voltage regulators operate by comparing the actual output voltage to some fixed reference voltage. Any difference is amplified and used to control the regulation element in such a way as to reduce the voltage error. This forms a negative feedback control loop; increasing the open-loop gain tends to increase regulation accuracy but reduce stability. (Stability is avoidance of oscillation, or ringing, during step changes.) There will also be a trade-off between stability and the speed of the response to changes. If the output voltage is too low (perhaps due to input voltage reducing or load current increasing), the regulation element is commanded, up to a point, to produce a higher output voltage—by dropping less of the input voltage (for linear series regulators and buck switching regulators), or to draw input current for longer periods (boost-type switching regulators); if the output voltage is too high, the regulation element will normally be commanded to produce a lower voltage. However, many regulators have over-current protection, so that they will entirely stop sourcing current (or limit the current in some way) if the output current is too high, and some regulators may also shut down if the input voltage is outside a given range .

Linear regulators are based on devices that operate in their linear region (in contrast, a switching regulator is based on a device forced to act as an on/off switch). Linear regulators are also classified in two types: series regulators, shunt regulators.

In the past, one or more vacuum tubes were commonly used as the variable resistance. Modern designs use one or more transistors instead, perhaps within an integrated circuit. Linear designs

have the advantage of very "clean" output with little noise introduced into their DC output but are most often much less efficient and unable to step-up or invert the input voltage like switched supplies. All linear regulators require a higher input than the output. If the input voltage approaches the desired output voltage, the regulator will "drop out". The input to output voltage differential at which this occurs is known as the regulator's drop-out voltage. Low-dropout regulators (LDOs) allow an input voltage that can be much lower (i.e., they waste less energy than conventional linear regulators).

Entire linear regulators are available as integrated circuits. These chips come in either fixed or adjustable voltage types. Examples of some integrated circuits are the 723 general purpose regulator and 78xx /79xx series. 7805 is a three terminal linear voltage regulator IC with a fixed output voltage of 5V which is useful in a wide range of applications. It is used to give a constant output voltage of 5V to the microcontroller to initialize the system.



Fig 3.5. VOLTAGE REGULATOR

3.1.6 IR SENSOR

An infrared (IR) sensor is an electronic device that measures and detects infrared radiation in its surrounding environment. Infrared radiation was accidentally discovered by an astronomer named William Herchel in 1800. While measuring the temperature of each color of light (separated by a prism), he noticed that the temperature just beyond the red light was highest. IR is invisible to the human eye, as its wavelength is longer than that of visible light (though it is still on the same electromagnetic spectrum). Anything that emits heat gives off infrared radiation. There are two types of infrared sensors: active and passive. Active infrared sensors both emit and detect infrared radiation. Active IR sensors have two parts: a light emitting diode (LED) and a receiver. When an object comes close to the sensor, the infrared light from the LED reflects off of the object and is detected by the receiver. Active IR sensors act as proximity sensors, and they are commonly used in obstacle detection systems.

Passive infrared (PIR) sensors only detect infrared radiation and do not emit it from an LED. Passive infrared sensors are comprised of:

- Two strips of pyroelectric material (a pyroelectric sensor)
- An infrared filter (that blocks out all other wavelengths of light)
- A fresnel lens (which collects light from many angles into a single point)
- A housing unit (to protect the sensor from other environmental variables, such as humidity)

PIR sensors are most commonly used in motion-based detection, such as in-home security systems. When a moving object that generates infrared radiation enters the sensing range of the detector, the difference in IR levels between the two pyroelectric elements is measured. The sensor then sends an electronic signal to an embedded computer, which in turn triggers an alarm. These IR sensors are used for line of sight object detection using which the system gets an input of the traffic density. Traffic density is measured as “low, medium and high”. Based on this density reading the system adjusts and varies the traffic signal duration for that particular way.

There are four sensors placed at four sides of a four way road which checks the density of the area covered by the sensors. As the vehicle passes these IR sensors, the IR sensor will detect the vehicle & will send the information to the microcontroller. The microcontroller will sense the density of vehicles and provide the glowing time to the LED as per the density of vehicles.

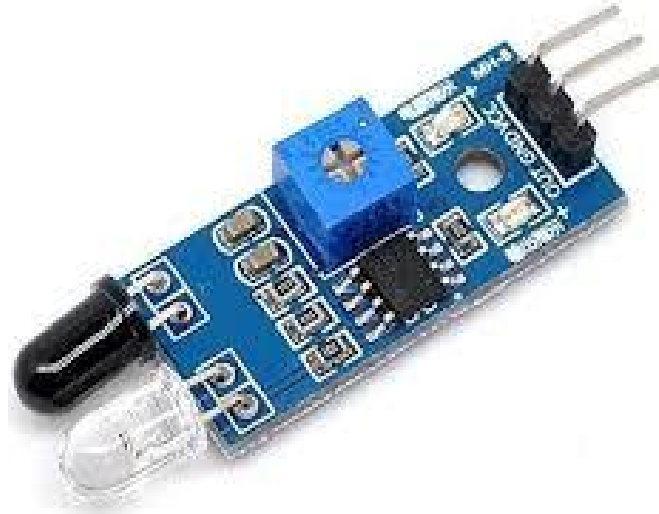


Fig 3.6. IR SENSOR

3.1.7.CONNECTING WIRES

(As per required)

3.2 BLOCK DIAGRAM

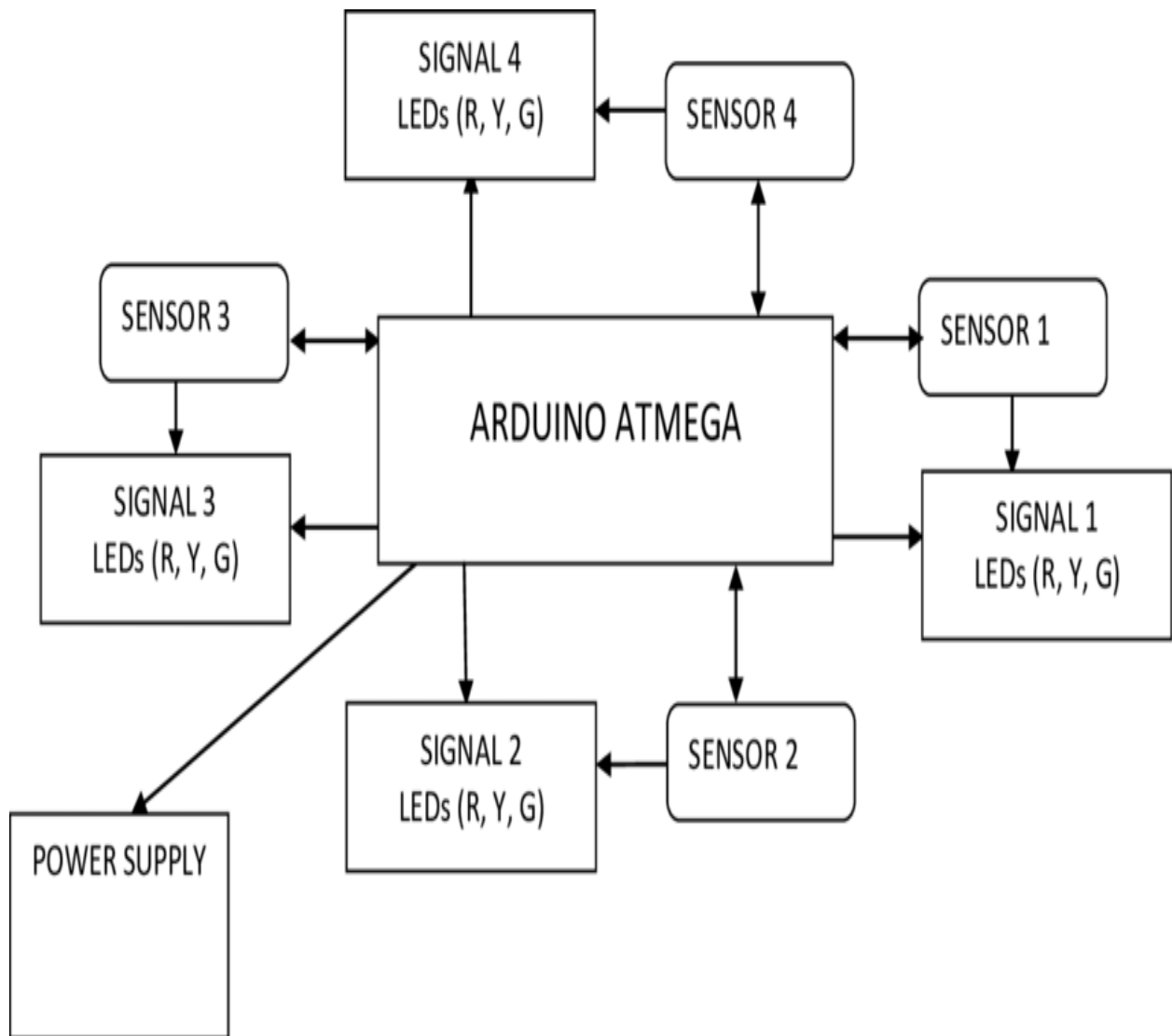


Fig 3.7.Block Diagram of Model

3.3 WORKING PRINCIPLE

The model works on the principle of changing of Traffic signals based on the density through an assigned section of the road. There are four sensors placed at four sides of a four way road which checks the density of the area covered by the sensors. Here we are using IR sensors to design an intelligent traffic control system. In order to measure the density of traffic on each side, IR sensors will be kept on either side of the road at a specific distance. Each of the IR sensors consists of an IR transmitter and an IR receiver. Just as the name suggests, the IR transmitter transmits the IR rays and the receiver is responsible to receive the rays. The whole system is controlled by the microcontroller which is the Arduino. Arduino is interfaced with Serial to parallel IC(74HC595) and IR sensors. As the vehicle passes through these IR sensors, the IR sensor will detect the vehicle & will send the information to the Arduino. The total no of IR sensors required are 4 and Led's 12. Three sets of LEDs via Green, Yellow and Red are used to indicate the GO state, Ready to Go state and WAIT state. The traffic signal will be tuned with a default timing of 10 seconds of green light and all other signals will be red. After 10 seconds two signals will be yellow for 4 seconds and another two will be red. This condition will be followed till all the IR sensors receiving the signals or all the IR sensors are not getting signals. The LEDs G (green), Y (yellow) and R (red) glow in the following sequence.

- G1-R2-R3-R4
- Y1-Y2-R3-R4
- R1-G2-R3-R4
- R1-Y2-Y3-R4
- R1-R2-G3-R4
- R1-R2-Y3-Y4
- R1-R2-R3-G4
- Y1-R2-R3-Y4

i.e., timing based traffic signals will be automatically implemented when all the signals have the same condition. When the condition changes, Let us suppose when the first side traffic signal is green and at that time third side traffic signal's IR sensor receives data then after the first traffic signal it will automatically shift towards the third traffic signal without moving to the second traffic signal.

- G1-R2-R3-R4
- Y1-R2-Y3-R4

- R1-R2-G3-R4

Similarly, Let green light is On in the fourth traffic signal for 20 seconds and during that time second traffic signal's IR sensor receiving data then after green light it will take 4 seconds delay for yellow light or we can say that the delay for pedestrians to walk in order to ensure their safety and then it will automatically shifts towards second traffic signal.

- R1-R2-R3-G4
- R1-Y2-R3-Y4
- R1-G2-R3-R4

Just taking into consideration the above conditions more further and let us suppose after second signal again forth signal's IR sensor receiving data then after 10 seconds and 4 seconds delay signal is green for forth lane.

- R1-G2-R3-R4
- R1-Y2-R3-Y4
- R1-R2-R3-G4

This circuit consists of 4 IR sensors, an atmega8 microcontroller, and 4 traffic lights.

The IR transmitter looks like an LED. This IR transmitter always emits IR rays from it. The operating voltage of this IR transmitter is 2 to 3v. These IR (infra red) rays are invisible to the human eye. But we can view these IR rays through a camera.

IR receiver receives IR rays that are transmitted by IR transmitter. Normally an IR receiver has high resistance in order of mega ohms, when it is receiving IR rays the resistance is very low. The operating voltage of the IR receiver is also 2 to 3V.

We have to place these IR pairs in such a way that when we place an obstacle in front of this IR pair, the IR receiver should be able to receive the IR rays. When we give the power, the transmitted IR rays hit the object and reflect back to the IR receiver.

Instead of traffic lights, you can use LEDs (RED, GREEN, YELLOW). In a normal traffic system, you have to light the LEDs on time basis. If the traffic density is high on any particular path, then glows green LED of that particular path glows and the red LEDs glow for the remaining paths.

In a normal traffic system, we allow the traffic for a time delay of 1 minute for each path.

NOTE:

The working of the project is divided into three steps:

1. If there is traffic at all the signals, then the system will work normally by controlling the signals one by one.
2. If there is no traffic near a signal, then the system will skip this signal and will move on to the next one. For example, if there is no vehicle at signal 2, 3 and currently the system is allowing vehicles at signal 1 to pass. Then after signal 1, the system will move on to signal 4 skipping signal 2 and 3.
3. If there is no traffic at all the 4 signals, then also the system will work normally by controlling the signals one by one.

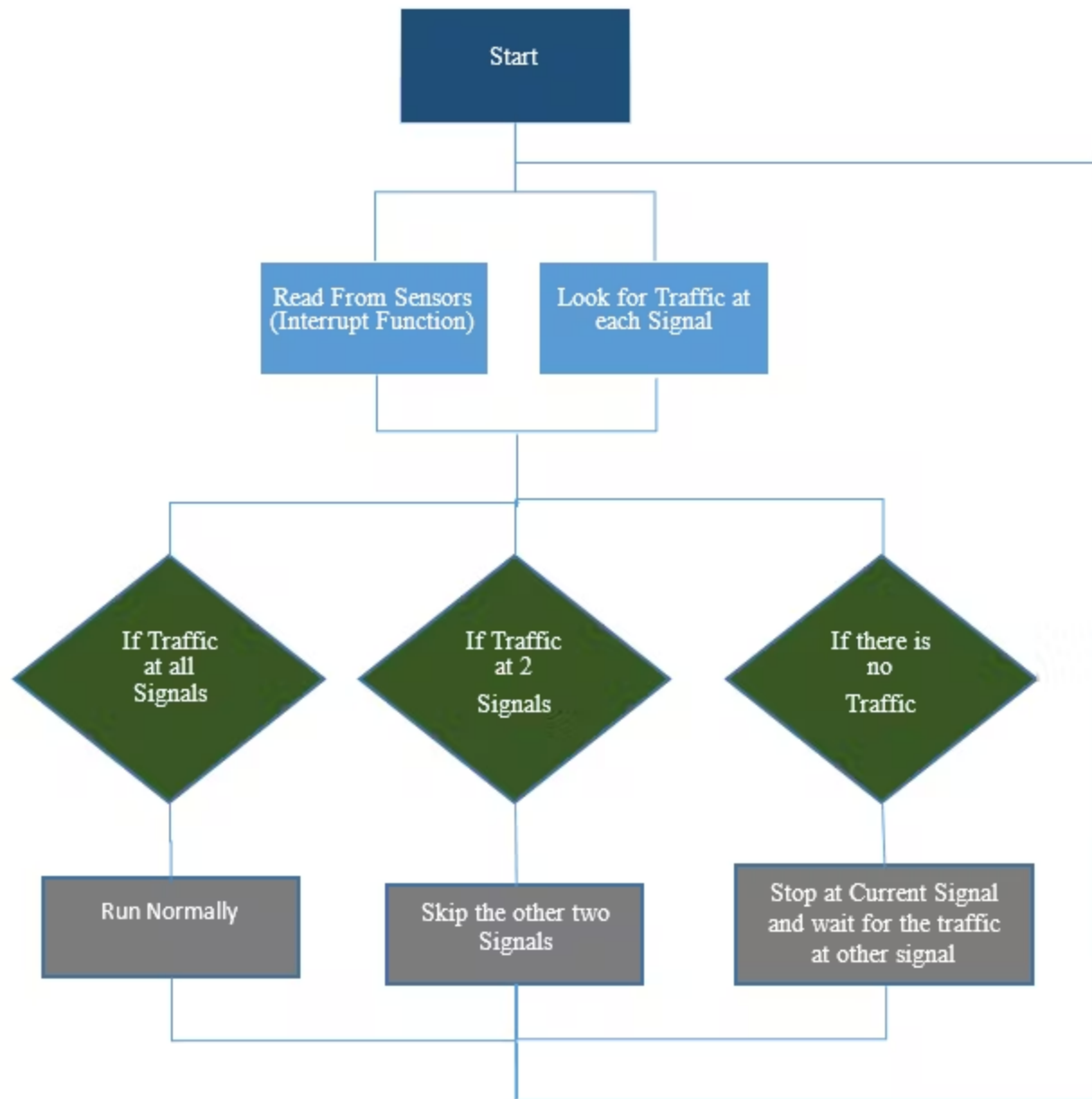


Fig 3.8. Flowchart of the Working

CHAPTER-4

RESULT ANALYSIS

Results include the successful operation of the traffic control and monitoring system. The system contains four IR transmitters and an IR receiver for traffic density measurement which are mounted on either side of roads respectively. The IR system gets activated whenever any vehicle passes on the road between IR sensors. When one sensor will be ON at that time density will be less when two sensors will be ON at that traffic density is medium when all 3 sensors will be ON at that time density will be high. Microcontroller controls the IR system and counts the number of vehicles passing on the road. Based on different vehicle counts, the microcontroller takes decisions and updates the traffic light delays as a result.

a) Case 1

In this case the density is highest on road 1 due to the presence of objects on road 1. The higher density will cause the green light on Road 1 to go green while the red light will occur for Road 2, road 3 & road 4.

b) Case 2

In this case density occurs at Road 2 resulting in the green light going green on road 2 and remaining road 1, road 3 and road 4 have red lights.

c) Case 3

In this case density occurs at road 3 resulting in the green light going green on road 3 and the red light occurs for road1, road 2 & road 4 respectively

d) Case 4

In this case density occurs at road 4 resulting in the green light to go green on road 4 and red light occurs for road 1, road 2 & road 3 respectively.

Limitations of the circuit:

- 1) Power Source: The traffic control system operates entirely on the power supplied from the battery or the power source.
Solution : We can use solar batteries equipped with solar panels as a backup power source.
- 2) Maintenance of IR Sensors: IR sensors are prone to damage as the key are exposed on the lanes 24*7 in all weather conditions.
Solution: Proper maintenance and frequent inspection of the sensors are required for their steady operation.

- 3) IR Sensors sometimes absorb normal light also, thus they work improperly.
Solution: The sensors must be aligned properly with ample distance present between them.
- 4) The density of the cars cannot be measured accurately for black color cars and cars present in between tall trucks and vehicles.
Solution: The sensors must be placed at various heights so that the error can be minimized.

ADVANTAGES:

1. Avoids wastage of time due to traffic.
2. Fully automatic.
3. Low power consumption.
4. Easy access to traffic lights.
5. Low cost to design
6. We can create many replicas with a microcontroller to control appliances.

CHAPTER - 5

FUTURE SCOPE AND CONCLUSION

5.1 FUTURE SCOPE

Due to continuous growth of population in the world, it is a great challenge for the upcoming generation to manage the traffic system. Much improvement will come in the future. To manage the conventional transport system we should think of an intelligent and automatic way of controlling the system. As the population increases, it increases the number of vehicles also. To control the huge number of vehicles, intelligent methods should be adopted. In the future we can use the image sensor or imager. It does it work by producing the image of the roads. It creates the image by converting the variable attenuation of light into a signal that conveys the picture. Imagers used both digital and analog electronics devices. The Future scope includes profiling the traffic by storing the data and managing the traffic lights according to the collected data. The Profiling can also be used for a traffic study and the variation in traffic density throughout the day, week, month, or year. Further, we can optimize this system for emergency Vehicles such as ambulances. The traffic data collected can be used to locate different routes for a specific daily vehicle to avoid the congestion problem.

5.2 CONCLUSION

In this research we have worked on the Congestion problem for such special areas which have dense traffic density. The system works on traffic-related problems such as traffic jams; unreasonable latency time of stoppage of vehicles, emergency vehicles or forcibly passing, etc can be solved. Using this system configuration we try to reduce the possibility of traffic jams, caused by traffic lights. A number of passing vehicles in the fixed time slot on the road decides the density range of traffic and on the basis of vehicle density calculation, microcontrollers decide the traffic light delays. There is an exigent need for an efficient traffic management system in our country, as India meets with 384 road accidents every day. To reduce this congestion and unwanted time delay in traffic an advanced system is designed here in this project. With field application of this technology, the maddening chaos of traffic can be effectively channelized by distributing the time slots based on the merit of the vehicle load in certain lanes of multi-junction crossing. We have successfully implemented the prototype at laboratory scale with remarkable outcome. The next step forward is to implement this schema as a real life scenario for first hand results, before implementing it on the largest scale. We believe that this may bring a revolutionary change in the traffic management system on its application in the actual field environment.

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